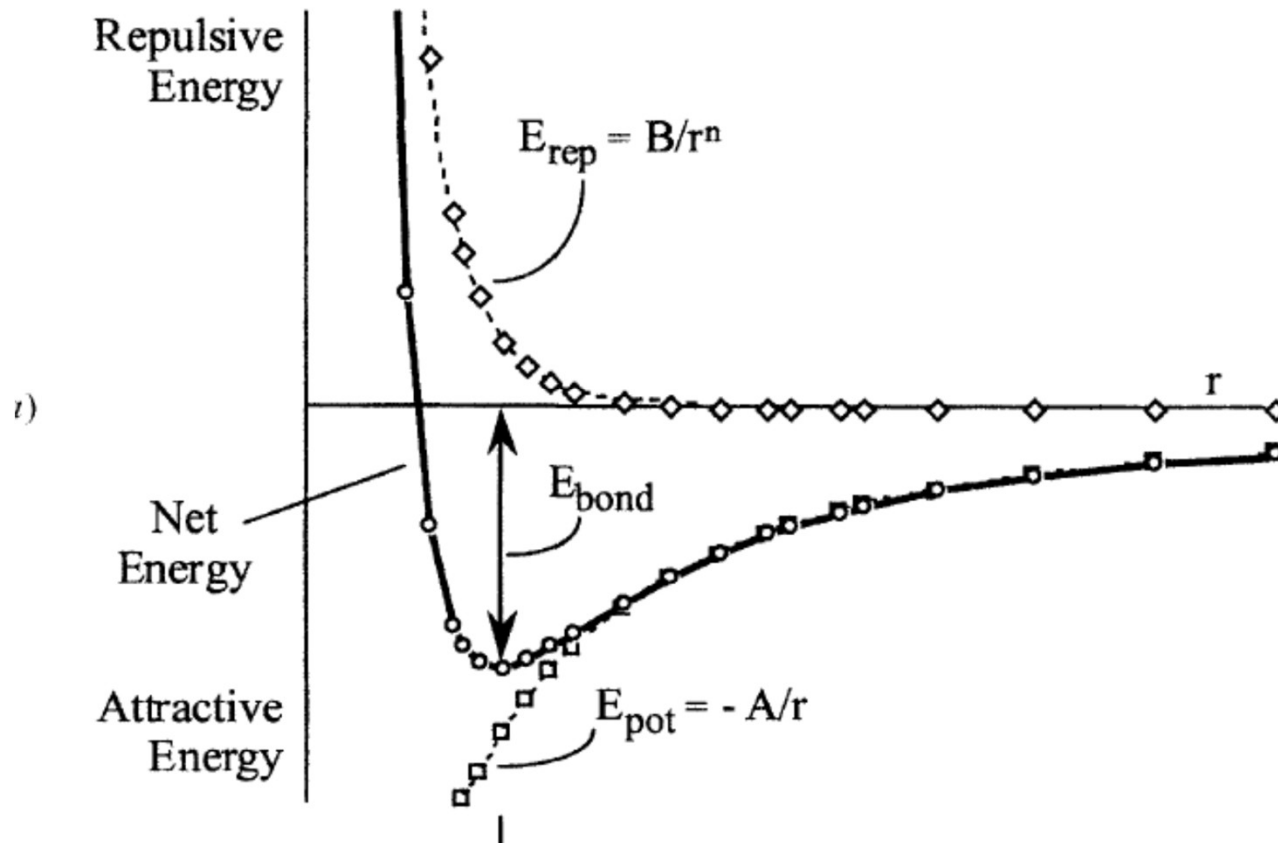


Structure of Ceramics

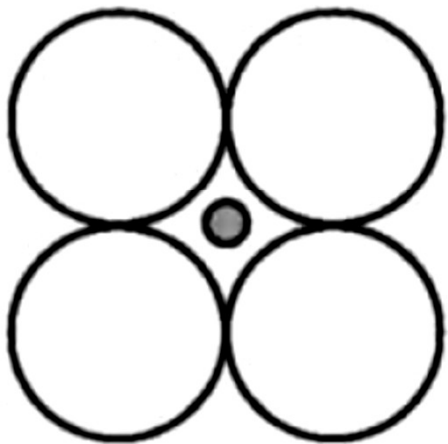
Pauling's rules

Reference: Barsoum, chapter 3, pp. 52 – 63
Hench & West, chapter 1, pp. 1-20

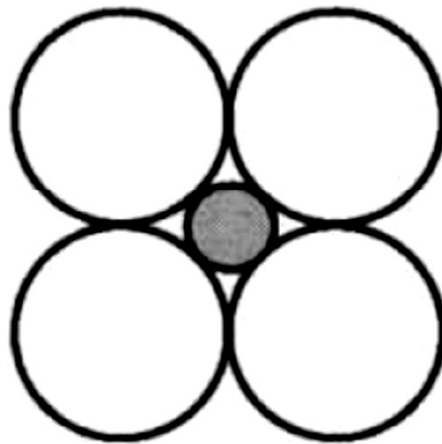
- The Na^+ and Cl^- ions form a lattice in 3D with multiple bonds
- The arrangement that leads to minimum E_{pot} is the most stable structure.
- The lattice is associated with a specific coordination number and lattice parameter.



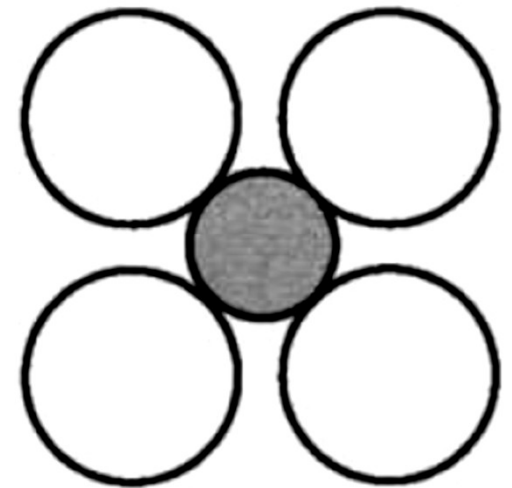
- **Radius ratio** = $r_{\text{cat}} / r_{\text{an}}$
- A radius ratio < than a critical value – leads “unstable” configuration
- All anions touch cation – to minimize the internal energy.
- Leads to an overlap of outer shells of anions → electrostatic repulsion (Born repulsion) and a higher energy configuration.
- A Minimum radius ratio is required to maintain a stable configuration in a given CN (here, CN of 4)



unstable



Critically stable



stable

Rule 1: “Maximum packing of anions around a reference cation”

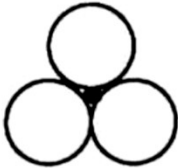
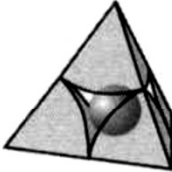
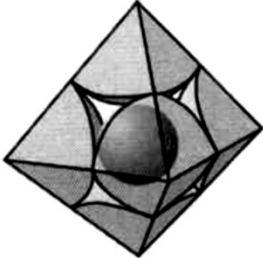
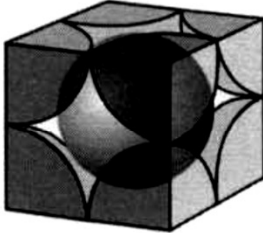
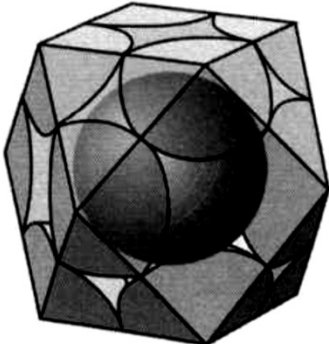
- **Coordination polyhedron of anions formed around the cation (why not vice versa)**
 - Anions are usually bigger than cations
 - Space filling is determined by arrangement of anions in space
 - Cations occupy their interstitial sites in an ordered manner.
- **Cation- anion distance is determined by their sum of radii (Empirical)**
- **Coordination number determined by the ratio of their radii**

CN = number of anions around a cation

$$\text{Radius ratio} = r_{\text{cat}} / r_{\text{an}}$$

CN only stable when $r_{\text{cat}} / r_{\text{an}}$ is greater than a critical value

Typical anion arrangements

Coordination number	Arrangement of ions around central ion	Range of cation/anion ratios	Structure
3	corners of a triangle	≥ 0.155	
4	corners of a tetrahedron	≥ 0.225	
6	corners of a octahedron	≥ 0.414	
8	corners of a cube	≥ 0.732	
12	corners of a cuboctahedron	≈ 1.000	

Layered structure of anions

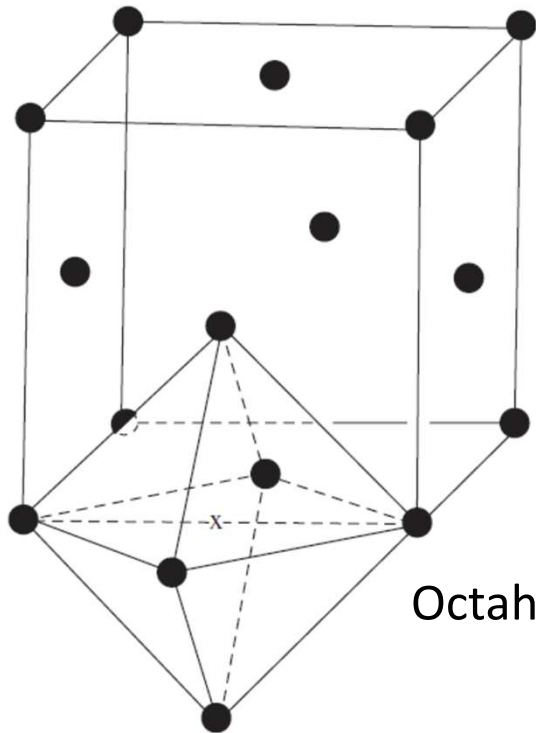
Tetrahedral voids in fcc or hcp of anions

Octahedral voids in fcc or hcp of anions

Centres of cubes in a simple cubic arrangement of anions

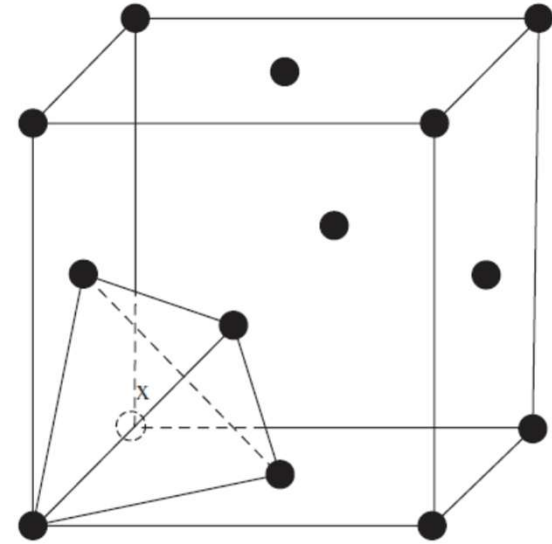
➤ Observed typically in Organic compounds, rarely in ceramics

Interstitial sites in fcc of anions



Octahedral sites

Tetrahedral sites



No of anions / unit cell = 4

No of octahedral sites = 4

No of tetrahedral sites = 8

Anions: octahedral = 1:1

anions: tetrahedral = 1:2

The most stable structure is usually the one with the maximum CN allowed by the radius ratio.

➔ The radius ratio $r_{\text{cat}}/r_{\text{an}}$ defines the upper limit of CN, not lower limit

Example, say radius ratio is 0.5

- Highest possible CN is 6 (>0.414 in slide 4)
- But lower coordinate numbers (2, 3, 4) are still possible.
- It is possible to form bonds with $r_o = r_{\text{cat}} + r_{\text{an}}$ even for CN of 2, 3, 4
- But the most stable structure will be the one with CN=6.

Why?

- Highest coordination leads to maximum number of bonds – maximum reduction of potential energy (formation of bond leads to release of energy).
- Pauling's first rule emphasizes coordination polyhedra of anions around cations.

“Maximum packing of oppositely charged ions around a reference ion”

List of ionic radii in Å

Ag ¹⁺	Al ³⁺	As ⁵⁺	Au ¹⁺	B ³⁺	Ba ²⁺	Be ²⁺	Bi ⁵⁺	Br ¹⁻	C ⁴⁺	Ca ²⁺	Cd ²⁺	Ce ⁴⁺
1.15	0.53	0.50	1.37	0.23	1.36	0.35	0.74	1.96	0.16	1.00	0.95	0.80
Cl ¹⁻	Co ²⁺	Co ³⁺	Cr ²⁺	Cr ³⁺	Cr ⁴⁺	Cs ¹⁺	Cu ¹⁺	Cu ²⁺	Dy ³⁺	Er ³⁺	Eu ³⁺	F ¹⁻
1.81	0.74	0.61	0.73	0.62	0.55	1.70	0.96	0.73	0.91	0.88	0.95	1.33
Fe ²⁺	Fe ³⁺	Ga ³⁺	Gd ³⁺	Ge ⁴⁺	Hf ⁴⁺	Hg ²⁺	Ho ³⁺	I ¹⁻	In ³⁺	K ¹⁺	La ³⁺	Li ¹⁺
0.77	0.65	0.62	0.94	0.54	0.71	1.02	0.89	2.20	0.79	1.38	1.06	0.74
Mg ²⁺	Mn ²⁺	Mn ⁴⁺	Mo ³⁺	Mo ⁴⁺	Na ¹⁺	Nb ⁵⁺	Nd ³⁺	Ni ²⁺	O ²⁻	P ⁵⁺	Pb ²⁺	Pb ⁴⁺
0.72	0.67	0.54	0.67	0.65	1.02	0.64	1.00	0.69	1.40	0.35	1.18	0.78
Rb ¹⁺	S ²⁻	S ⁶⁺	Sb ³⁺	Sc ³⁺	Se ²⁺	Se ⁶⁺	Si ⁴⁺	Sm ²⁺	Sn ²⁺	Sn ⁴⁺	Sr ²⁺	Ta ⁵⁺
1.49	1.84	0.30	0.61	0.73	1.98	0.42	0.40	0.96	0.93	0.69	1.16	0.64
Te ²⁻	Te ⁶⁺	Th ⁴⁺	Ti ²⁺	Ti ⁴⁺	Tl ¹⁺	Tl ³⁺	U ⁴⁺	U ⁵⁺	V ²⁺	V ⁵⁺	W ⁴⁺	W ⁶⁺
2.21	0.56	1.00	0.86	0.61	1.50	0.88	0.97	0.76	0.79	0.54	0.65	0.58
Y ³⁺	Yb ³⁺	Zn ²⁺	Zr ⁴⁺									
0.89	0.86	0.75	0.72									

- Compare Mg²⁺ and O²⁻
- O twice as bigger as Mg

RR ~ 0.5 → CN of 6

MgO forms the Rock salt structure

2nd rule: electrical neutrality should be preserved in the crystal

- cation: anion ratio/ unit cell should follow stoichiometry
- Overall, the sum of positive and negative charges should be balanced within the unit cell.
- Consider M – metal ion, X – non-metal / metalloid ion
- Consider cubic packing of anions (either simple cubic or fcc). Ignore hcp for time being.
 - MX type of compounds can form Rock salt structure/ CsCl structure /Zinc Blende structure – Ratio of anion: cation sites = 1
 - MX₂ can form Fluorite structure – Ratio of anion: cation sites = 2
 - M₂X can form Antifluorite structure – Ratio of anion: cation sites = 0.5
- Exact choice of structure further depends on the type of bonding and the preferred CN.

If a compound is MX, it could form the following structures

- NaCl (CN 6 – octahedral voids filled in fcc)
- or Zinc Blende (CN 4 – half the tetrahedral voids filled in fcc)
- or CsCl structure (CN 8 – body centred positions filled in simple cubic)

Larger is the radius of the cation – more is the possibility of formation of CN = 8.

Exact structure preferred in nature is decided by a combination of

- Maximum CN allowed (Rule 1)
- Stoichiometry or electrical neutrality (Rule 2)

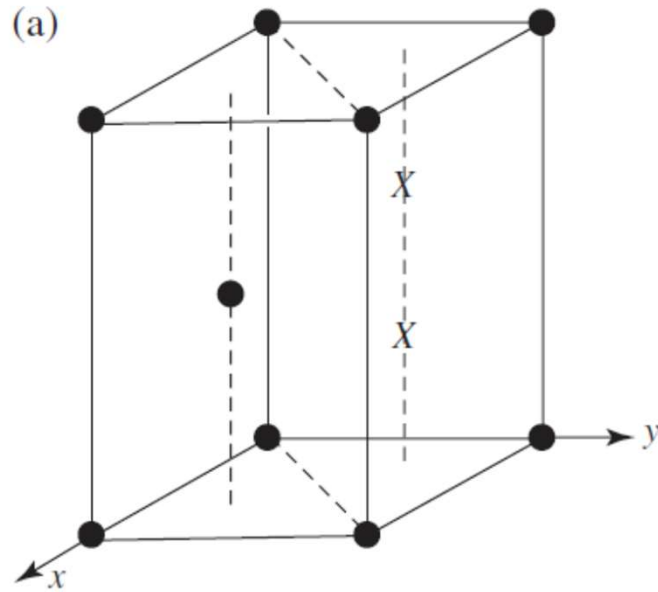
Other factors may be allowed in addition to Rule 1 and Rule 2

- **Propensity for covalency** leads to a preference for tetrahedral coordination.
- (Bonding in ceramics is ionic – covalent → partial charge transfer at the atomic level.)
- Elements that form sp^3 hybridization tend to stabilize tetrahedral coordination – sp^3 bonds are directional bonds.
- For example, MX with $RR = 0.5$ may still form ZnS structure (although preferred Cn is 6) if the bond is predominantly covalent.
- Cations with high polarizing power combined with anions with a high polarizability tend to form tetrahedral configuration

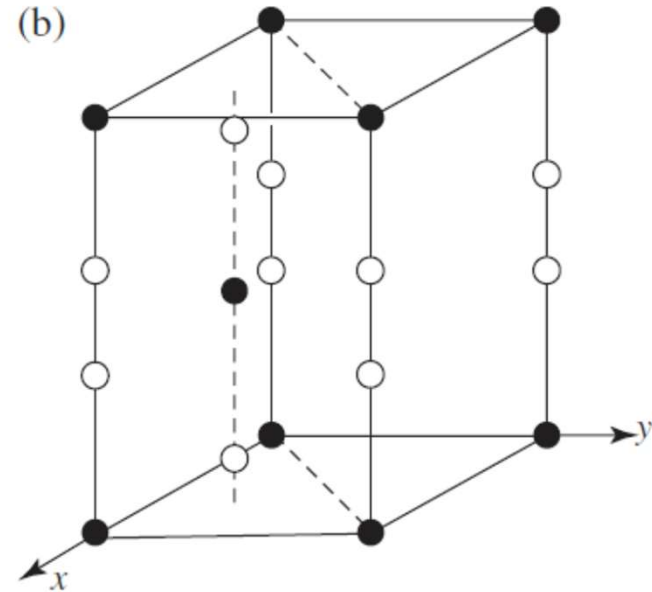
Octahedral sites

$$\left(\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right)$$

$$\left(\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right)$$



Tetrahedral sites



$$\left(0, 0, \frac{3}{8}\right)$$

$$\left(0, 0, \frac{5}{8}\right)$$

$$\left(\frac{2}{3}, \frac{1}{3}, \frac{7}{8}\right)$$

$$\left(\frac{2}{3}, \frac{1}{3}, \frac{1}{8}\right)$$

Number per hcp unit cell:

Atoms / anions: 2

Octahedral sites: 2

Tetrahedral sites: 4

MX

MX₂

M₂X

Common ceramic structures

(named after the mineral in which it was first observed).

Red – fcc of anions

Blue – hcp of anions

Green – simple

cubic of anions

Structure name	Anion packing	Coordination no. of M and X	Sites occupied by cations	Examples
Binary compounds				
Rock salt	Cubic close-packed	6 : 6 MX	All oct.	NaCl, KCl, LiF, KBr, MgO, CaO, SrO, BaO, CdO, VO, MnO, FeO, CoO, NiO
Rutile	Distorted cubic close-packed	6 : 3 MX ₂	1/2 oct.	TiO ₂ , GeO ₂ , SnO ₂ , PbO ₂ , VO ₂ , NbO ₂ , TeO ₂ , MnO ₂ , RuO ₂ , OsO ₂ , IrO ₂
Zinc blende	Cubic close-packed	4 : 4 MX	1/2 tet.	ZnS, BeO, SiC
Antifluorite	Cubic close-packed	4 : 8 M ₂ X	All tet.	Li ₂ O, Na ₂ O, K ₂ O, Rb ₂ O, sulfides
Wurtzite	Hexagonal close-packed	4 : 4 MX	1/2 tet.	ZnS, ZnO, SiC, ZnTe
Nickel arsenide	Hexagonal close-packed	6 : 6 MX	All oct.	NiAs, FeS, FeSe, CoSe
Cadmium iodide	Hexagonal close-packed	6 : 3 MX ₂	1/2 oct.	CdI ₂ , TiS ₂ , ZrS ₂ , MgI ₂ , VBr ₂
Corundum	Hexagonal close-packed	6 : 4 M ₂ X ₃	2/3 oct.	Al ₂ O ₃ , Fe ₂ O ₃ , Cr ₂ O ₃ , Ti ₂ O ₃ , V ₂ O ₃ , Ga ₂ O ₃ , Rh ₂ O ₃
CsCl	Simple cubic	8 : 8 MX	All cubic	CsCl, CsBr, CsI
Fluorite	Simple cubic	8 : 4 MX ₂	1/2 cubic	ThO ₂ , CeO ₂ , UO ₂ , ZrO ₂ , HfO ₂ , NpO ₂ , PuO ₂ , AmO ₂ , PrO ₂
Silica types	Connected tetrahedra	4 : 2 MO ₂	—	SiO ₂ , GeO ₂

FCC of anion based structures

Rock salt

Zinc blende

Antifluorite

Spinel

Rutile

HCP of anion based

Wurtzite

Corundum

Ilmenite

SC of anion based

Fluorite

CsCl

Other

Perovskite